1 Fig. 1.1 shows a model wind turbine with blades, each of length L, placed in moving air.

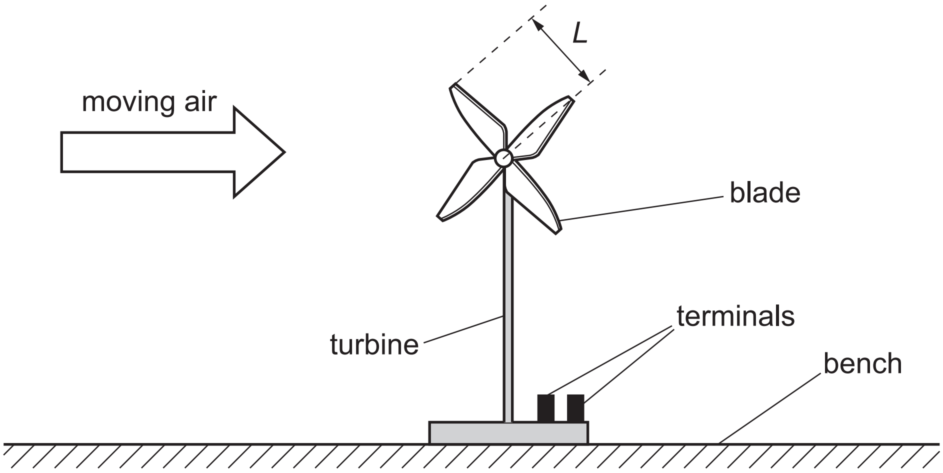

Fig. 1.1

The area of the circle swept by the blades of the turbine is A.

The output of the turbine has two terminals. The turbine is connected to a resistor of resistance R. At a speed v of the moving air, the current in the resistor is I.

The atmospheric pressure is P and the thermodynamic temperature of the air is T.

It is suggested that I is related to v by the relationship

$$\frac{I^2R}{Q} = \frac{APv^3}{2T}$$

where Q is a constant.

Plan a laboratory experiment to test the relationship between I and v.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for Q.

In your plan you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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Diagram

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